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Part I

Theoretical Background

Chapter 1

Mori-Zwanzig theory

The Mori-Zwanzig theory allows to observe the evolution of a dynamical system through the projection on a subset of its degrees of freedom, this is done through *projection operator techniques*.

The *projection operator technique* are powerful tools for the study of irreversible processes in many-body systems [5]. These tools has been introduced into statistical physics by S. Nakajima in 1958 and R. Zwanzig in 1960. The former to study the quantum statistical mechanics of steady diffusion [12], and the latter to give a new formulation of methods introduced in the statistical-mechanical theory of irreversibility [17]. In particular R. Zwanzig has developed the technique into a powerful tool for deriving formally accurate motion equations for classical or quantal probability densities [18, 19]. Later in 1965, H. Mori proposed a different tool, which leads to the linear *generalized Langevin equation* (GLE), an equation of the form

$$\frac{\mathrm{d}A}{\mathrm{d}t} = i\Omega A(t) - \int_0^t \mathrm{d}\tau K(\tau)A(t-\tau) + F(t) \tag{1.1}$$

where Ω, $K(t)$ and $F(t)$ are well-defined functions, called the frequency, memory function, and random force respectively. This equation describes the non-equilibrium evolution of any set of functions $A(t)$ defined on the phase-space of the arbitrary microscopic system, close to thermal equilibrium [11]. To now these variables $A(t)$ are called collective variables, relevant variables, coarse-grained (CG) variables or collective modes, depending on the context. Some years later in 1973 R. Zwanzig derived a class of generalized Langevin equations describing the motion of a system interacting with a heat bath, this GLE is exact but differs from that obtained by Mori's scheme for two reasons: for the presence of an arbitrary nonlinear force and for the fact that its validity is not restricted to small deviations from the thermal equilibrium [20]. In recent years D. Kauzlarić et all showed, in their work on the calculation of the friction matrix in the coarse-grained description of an atomistic system, that Mori GLE can be obtained from the Zwanzig's one if the system is close to the equilibrium and under the Markovian assumption, i.e. that the relevant variables (the functions $A(t)$ in the equation (1.1)) evolve in a time scale much larger than the memory time scale [8]. These techniques are is know in various context as well as in transport theory. For example in the dynamic light scattering or in the

liquids theory, the GLE is used to compute time-correlation functions under circumstances where the usual phenomenological equations do not apply. Indeed through the equation (1.1) can be obtained an equation for the time correlation function $C(t)$:

$$\frac{\mathrm{d}C(t)}{\mathrm{d}t} = i\Omega\, C(t) - \int_0^t \mathrm{d}\tau K(\tau)C(t-\tau)\,. \tag{1.2}$$

This is called the *memory-function equation* and the random force appears implicitly in $K(t)$ [1, 6].

If the dynamical system considered is Hamiltonian, in the Mori-Zwanzig theory we can use two equivalent approaches. The former applying the projection operator in a Hilbert space of probability densities function (pdf) to the Liouville equation, obtained the generalized Fokker-Planck and master equations [9, 17, 18, 19]. While the latter, applying the projection operator in the Liouville space, see appendix B, to the equation of motion of time-dependent property phase function, obtained GLE and transport equation [3, 7, 11, 13, 20]. The mathematical equivalence of these two approaches is given by the duality between Liouville space and the space of densities of states. In what follows, we choose to follow the second approach since this results in a more physically-intuitive derivation and we will derive the GLE for the CG variables with the *Zwanzig projection* and the *Mori projection*, respectively.

We assume that at the microscopic level the system is well described by classical mechanics and is composed by M^{FG} particles. We will refer to it as the *fine-grained* (FG) system. The microscopic state is characterised by variable $\boldsymbol{\gamma}(t)$ in $6M^{FG}$-dimensional phase-space. To derive a reduced description of this system, we consider a smaller set of phase function $\boldsymbol{\Gamma}(\boldsymbol{\gamma})$, which we call observables[1]. The time evolution of a function of phase variables is given by

$$\frac{\mathrm{d}\boldsymbol{\Gamma}(\boldsymbol{\gamma}(t))}{\mathrm{d}t} = \mathfrak{L}\,\boldsymbol{\Gamma}(\boldsymbol{\gamma}(t))\,, \quad \boldsymbol{\Gamma}(\boldsymbol{\gamma}(0)) = \boldsymbol{\Gamma}(\boldsymbol{\gamma}_0)\,. \tag{1.3}$$

Here, the operator $\mathfrak{L} \equiv iL$ is the *Liouvillian*, see appendix B, and the formal solution is given by

$$\boldsymbol{\Gamma}(\boldsymbol{\gamma}(t)) = \mathbf{e}^{\mathfrak{L}t}\,\boldsymbol{\Gamma}(\boldsymbol{\gamma}_0)\,. \tag{1.4}$$

This solution implicitly depends upon the full evolution of the FG system, therefore this equation does not provide a self-consistent evolution equation for the observables.

The Mori-Zwanzig formalism allows us to replace equation (1.3) for the observables by an equation of motion which depends on the microstate $\boldsymbol{\gamma}$ only through the CG variables themselves leading to a closed equation for the CG variables. For example, with the *Zwanzig projection* this is done by projecting the equation for the observables $\boldsymbol{\Gamma}$ onto the space of functions of the CG variables, which may be regarded as a subspace of the FG observable space[2].

[1] Throughout the present section, these observables could be quite general, but we will later apply the derivation made here to the particular choice

$$\boldsymbol{\Gamma}(\boldsymbol{\gamma}) = (\boldsymbol{\Gamma}_{\mathbf{R}}(\boldsymbol{\gamma}), \boldsymbol{\Gamma}_{\mathbf{P}}(\boldsymbol{\gamma})) = (\mathbf{R}_1(\boldsymbol{\gamma}), \ldots, \mathbf{R}_{M^{CG}}(\boldsymbol{\gamma}), \mathbf{P}_1(\boldsymbol{\gamma}), \ldots, \mathbf{P}_{M^{CG}}(\boldsymbol{\gamma}))\,,$$

corresponding to the positions and momenta of particles in the CG system composed of M^{CG} particles.

[2] This is true as long as some very broad measurability assumptions on the CG variables chosen are met; in practice, choosing CG variables which are continuous and differentiable in the FG variables is already sufficient, and this particular choice is implicitly assumed in the derivation presented here. In theory however, much more general choices can be made.

1.1 Zwanzig approach

1.1.1 The Zwanzig projection

We suppose that the state of the system is described by the canonical equilibrium distribution[3] $\rho^{eq}(\boldsymbol{\gamma})$, therefore the probability equilibrium measure is given by the Gibbs measure on the FG phase-space $X = \mathbb{R}^{6M^{FG}}$:

$$\mathbb{P}(\mathrm{d}\boldsymbol{\gamma}) = \rho^{eq}(\boldsymbol{\gamma})\mathrm{d}\boldsymbol{\gamma} = \frac{1}{\mathcal{Z}}\mathrm{e}^{-\beta\mathcal{H}(\boldsymbol{\gamma})}\mathrm{d}\boldsymbol{\gamma}\,. \tag{1.5}$$

We denote the *Zwanzig projection* as $\mathcal{P}_{\boldsymbol{\Gamma}}$:

$$\mathcal{P}_{\boldsymbol{\Gamma}}F(\boldsymbol{\gamma}) = \frac{1}{\Omega(\boldsymbol{\Gamma})}\int F(\boldsymbol{\gamma}')\,\delta\left(\boldsymbol{\Gamma}(\boldsymbol{\gamma}') - \boldsymbol{\Gamma}\right)\rho^{eq}(\boldsymbol{\gamma}')\mathrm{d}\boldsymbol{\gamma}' \tag{1.6}$$

where $F(\boldsymbol{\gamma})$ is a generic function acting on the space-phase and the normalisation factor, also called the *structure function*, represents the total volume of the surface $S(\boldsymbol{\Gamma})$ in phace-space, which is defined by $\boldsymbol{\Gamma}(\boldsymbol{\gamma}) = \boldsymbol{\Gamma}$:

$$\Omega(\boldsymbol{\Gamma}) = \int \delta\left(\boldsymbol{\Gamma}(\boldsymbol{\gamma}') - \boldsymbol{\Gamma}\right)\mathbb{P}(\mathrm{d}\boldsymbol{\gamma}')\,. \tag{1.7}$$

In other words, *structure function* represents the number of micro-states compatible with the macro-state $\boldsymbol{\Gamma}(\boldsymbol{\gamma}) = \boldsymbol{\Gamma}$ [7].

The projection operator $\mathcal{P}_{\boldsymbol{\Gamma}}$ selects from an arbitrary function $F(\boldsymbol{\gamma})$ that part, $\mathcal{P}_{\boldsymbol{\Gamma}}F(\boldsymbol{\gamma})$, which depends on $\boldsymbol{\gamma}$ only through $\boldsymbol{\Gamma}(\boldsymbol{\gamma})$. For a given $\boldsymbol{\gamma}$ we determine the observables $\boldsymbol{\Gamma} = \boldsymbol{\Gamma}(\boldsymbol{\gamma})$. The integral in the numerator represents the amount of $F(\boldsymbol{\gamma})$ contained in the correspondence surface $S(\boldsymbol{\Gamma})$, while the *structure function* in the denominator $\Omega(\boldsymbol{\Gamma}(\boldsymbol{\gamma})) = \Omega(\boldsymbol{\Gamma})$ allows us to spread this amount on the surface $S(\boldsymbol{\Gamma})$. Therefore, loosely speaking, $\mathcal{P}_{\boldsymbol{\Gamma}}$ "average" the function $F(\boldsymbol{\gamma})$ over the surface $S(\boldsymbol{\Gamma}(\boldsymbol{\gamma}))$. The fact that $\mathcal{P}_{\boldsymbol{\Gamma}}$ is a projection operator can be easily verified by showing that it satisfies the necessary and sufficient condition for a projection operator, $\mathcal{P}_{\boldsymbol{\Gamma}}^2 = \mathcal{P}_{\boldsymbol{\Gamma}}$ [18].

1.1.2 Equivalence of Zwanzig projection and conditional expectation

We will now show that this projection is equivalent to taking the conditional expectation:

$$\mathcal{P}_{\boldsymbol{\Gamma}}F \equiv \mathbb{E}\left[F|\,\boldsymbol{\Gamma}\right] = \langle F\rangle_{\boldsymbol{\Gamma}}$$

with respect to the equilibrium distribution, for a generic function $F = F(\boldsymbol{\gamma})$ acting on the phase space and for a specific value of the CG variables $\boldsymbol{\Gamma}(\boldsymbol{\gamma}) = \boldsymbol{\Gamma}$ [2, 3]. This is done because it will be useful for the following sections.

Let $A \in L^2(X, \mathbb{P})$, i.e. a phase function in the Liouville space (see appendix B). The observed value of the phase function A is the expected value $\mathbb{E}[A]$, defined by (A.1.9), with respect to the Gibbs measure (1.5). Therefore, $A(\boldsymbol{\gamma})$ may be considered a random variable, see definition (A.1.7). We now focus our attention on a set of functions

$$\left\{A_I \in L^2(X, \mathbb{P}):\ I = 1, \ldots, M^{CG} \ \text{ and } \ M^{CG} \leq M^{FG}\right\},$$

[3] Different choices are possible. For example it is also perfectly possible to carry out the same derivation using the micro-canonical equilibrium measure.

i.e. on the set of CG variables, which define a space $V = \mathbb{R}^{M^{CG}}$.

We are now in position to define the projection operator. Given a phase function B, we define the projection operator over the CG variables as

$$\mathcal{P}_{\mathbf{a}} B = \int_{A^{-1}(\{\mathbf{a}\})} B(\boldsymbol{\gamma}) \mathbb{P}_{\mathbf{a}}(\mathrm{d}\boldsymbol{\gamma}) \,,$$

where $A^{-1}(\{\mathbf{a}\}) = \{\boldsymbol{\gamma} \in X : \; A(\boldsymbol{\gamma}) = \mathbf{a}\}$ is the preimage of $\boldsymbol{\Gamma} \in \mathbb{R}^{M^{CG}}$.

1.1.3 Generalized Langevin Equation

Differentiating both sides of the relation (1.4) with respect to time we can rewrite the equation (1.3) as

$$\frac{\mathrm{d}\boldsymbol{\Gamma}(\boldsymbol{\gamma}(t))}{\mathrm{d}t} = \mathfrak{L}\, \mathbf{e}^{\mathfrak{L}t}\, \boldsymbol{\Gamma}(\boldsymbol{\gamma}_0) = \mathbf{e}^{\mathfrak{L}t} \mathfrak{L}\, \boldsymbol{\Gamma}_0 \,, \tag{1.8}$$

where $\boldsymbol{\Gamma}_0 = \boldsymbol{\Gamma}(\boldsymbol{\gamma}_0)$.

1.2 Mori approach

Chapter 2

Canonical Mapping and Coarse-Graining

Chapter 3

Numerical Method

3.1 Langevin Dynamics Integrator: BAOAB method

In this work we simulated the Langevin Dynamics by the BAOAB method, which was derived and analysed by B. Leimkuhler and C. Matthews [10].

The aim of their work is to compare Langevin dynamics integrators in term of their invariant distributions and demonstrate a superconvergence property of a certain method in the high friction limit (or in the null mass limit). Numerically they implemented the various methods considered and compared the accuracy of configurational sampling for different values of the friction parameter γ and a range of timesteps.

Their results show that the best method to sample the configurational Gibbs-Boltzmann distribution[1] is the BAOAB method and in the context of molecular dynamics it is a cheap and easy to implement scheme using only a single force vector per timestep:

$$
\begin{aligned}
p_{n+1/2} &= p_n - \frac{1}{2}\nabla U\left(x_n\right)\, dt\,; \\
x_{n+1/2} &= x_n + \frac{1}{2}\, M^{-1}\, p_{n+1/2}\, dt\,; \\
\hat{p}_{n+1/2} &= c_1\, p_{n+1/2} + c_2\, M^{1/2}\, R_{n+1}\,; \\
x_{n+1} &= x_{n+1/2} + \frac{1}{2}\, M^{-1}\, \hat{p}_{n+1/2}\, dt\,; \\
p_{n+1} &= \hat{p}_{n+1/2} - \frac{1}{2}\nabla U\left(x_{n+1}\right)\, dt\,.
\end{aligned}
$$

Here, R_i is an N-vector of iid normal random numbers with mean 0 and variance 1, M is the diagonal mass matrix, dt is the timestep,

$$
c_1 = e^{-\tilde{\gamma}\, dt} \quad \text{and} \quad c_2 = \sqrt{\left(1 - c_1^2\right)/\beta}
$$

are evaluated for a given friction parameter $\tilde{\gamma} = \gamma\, M^{-1}$ and a given temperature parameter $\beta = (K_B T)^{-1}$, where K_B is the Boltzmann's constant.

[1] This means the calculation of functions averages of phase space variables of a molecular N-body system modelled at constant temperature.

3.1.1 An idea of how the BAOAB method was derived

Background

An Itô stochastic differential equation (SDE) can be written as

$$d\mathbf{x}(t) = \mathbf{a}\left[\mathbf{x}(t), t\right] dt + \mathbf{b}\left[\mathbf{x}(t), t\right] d\mathbf{W}(t)\,,$$

where $\mathbf{x}$ are the n stochastic process, $\mathbf{a}(\mathbf{x}, t)$ and $\mathbf{b}(\mathbf{x}, t)$ are certain know functions, $d\mathbf{W} \equiv \boldsymbol{\xi}(t)\, dt$ is an n variable Wiener process and $\boldsymbol{\xi}(t)$ is the white noise. This SDE corresponds an Fokker-Planck equation if the stochastic process described by a conditional probability density $p(\mathbf{x}, t \,|\, \mathbf{x_0}, t_0) \equiv p$ satisfy

$$\partial_t\, p = \mathscr{L}(\mathbf{x}, t)\, p = -\sum_i \partial_i \left[A_i(\mathbf{x}, t)\, p\right] + \frac{1}{2}\sum_{ij} \partial_i \partial_j \left[B_{ij}(\mathbf{x}, t)\, p\right]$$

where $\mathscr{L}(\mathbf{x}, t)$ is a second-order differential operator and the coefficients of RHS are such that $\mathbf{A}(\mathbf{x}, t) = \mathbf{a}(\mathbf{x}, t)$ and $\underline{\mathbf{B}}(\mathbf{x}, t) = \mathbf{b}(\mathbf{x}, t)\mathbf{b}^T(\mathbf{x}, t)$. The Fokker-Planck equation describes diffusion processes, the vector $\mathbf{A}(\mathbf{x}, t)$ is know as drift vector and the matrix $\underline{\mathbf{B}}(\mathbf{x}, t)$ is know as the diffusion matrix. From its definition this matrix is positive semidefinite and symmetric [4].

Let us consider an Orstein-Uhlenbeck (OU) SDE, that is, an Itô equation where the coefficients are

$$\mathbf{a}(\mathbf{x}) = -k\,\mathbf{x} \quad \text{and} \quad b = \sqrt{D}$$

here k and D are positive parameters. The solution of one dimensional OU-SDE can be computed by the Itô's formula applied to $y = xe^{k\,t}$[4]. We get

$$x(t) = x_0(t_0)\, e^{-k(t-t_0)} + \sqrt{D}\int_{t_0}^{t} dW(t') e^{-k(t-t')}\,,$$

and the average and variance are respectively:

$$\mathbb{E}[x(t)] = \mathbb{E}[x_0(t_0)]\, e^{-k(t-t_0)}\,,$$
$$\mathbb{E}\left[\left(x(t) - \mathbb{E}[x(t)]\right)^2\right] = \left[Var(x_0) - \frac{D}{2k}\right] e^{-2k(t-t_0)} + \frac{D}{2k}\,.$$

We can easily show in the stationary limit ($t \gg t_0$) and applying the equipartition theorem that the diffusive parameter D is equal to $(2k)/\beta$.

Langevin Dynamics

Langevin dynamics combines a conservative dynamic with the OU stochastic vector field as a forcing for the momentum, so we are assuming that the stochastic force is not affected by the potential:

$$M\frac{d^2\mathbf{x}}{dt^2} = -\boldsymbol{\nabla} U(\mathbf{x}) - \gamma\,\frac{d\mathbf{x}}{dt} + \sqrt{(2\gamma)/\beta}\,\boldsymbol{\xi}(t)$$

where we introduced physical quantities: M is the mass of the particle, $\mathbf{x}$ is the coordinate of position, γ is the friction parameter and $\beta = (K_B T)^{-1}$ is the temperature parameter. We can rewrite the equation as

$$\begin{cases} d\mathbf{x} = \mathbf{v}\, dt \\ M d\mathbf{v} = \left[-\boldsymbol{\nabla} U(\mathbf{x}) - \gamma\,\mathbf{v}\right]\, dt + \sqrt{(2\gamma)/\beta}\, d\mathbf{W} \end{cases}$$

and for the canonical coordinates $\{\mathbf{q}(t), \mathbf{p}(t)\}$ we get

$$\begin{cases} d\mathbf{q} = M^{-1}\,\mathbf{p}\,dt \\ d\mathbf{p} = [-\boldsymbol{\nabla} U(\mathbf{q}) - \tilde{\gamma}\,\mathbf{p}]\;dt + \sqrt{(2\tilde{\gamma})/\beta}\,M^{1/2} d\mathbf{W} \end{cases}$$

where $\tilde{\gamma} = \gamma/M$. From the system of two first-order equations can we compute the drift and diffusion coefficients

$$\begin{aligned} (A_{\mathbf{q}})_i &= M^{-1}\,p_i\,; & (B_{\mathbf{qq}})_{ij} &= (B_{\mathbf{pq}})_{ij} = 0\,; \\ (A_{\mathbf{p}})_i &= (-\boldsymbol{\nabla} U(\mathbf{q}) - \tilde{\gamma}\,\mathbf{p})_i\;; & (B_{\mathbf{pp}})_{ij} &= \frac{2\tilde{\gamma}}{\beta}\,M\,\delta_{ij}\,; \end{aligned}$$

and thus determine the Fokker-Planck operator for the Langevin dynamics

$$\begin{aligned} \mathscr{L}_{LD}\,\varphi = &-M^{-1}\sum_i p_i\,\partial_{q_i}\,\varphi \;+\; \boldsymbol{\nabla} U(\mathbf{q})\sum_i \partial_{p_i}\,\varphi \;+ \\ &+\;\tilde{\gamma}\sum_i \partial_{p_i}\,p_i\,\varphi \;+\; \tilde{\gamma}\sum_i p_i\,\partial_{p_i}\,\varphi \;+\; \tilde{\gamma}\,K_B T\sum_i M\,\partial_i^2\,\varphi = \\ &= -M^{-1}\mathbf{p}\cdot\boldsymbol{\nabla}_q\,\varphi \;+\; \boldsymbol{\nabla} U(\mathbf{q})\cdot\boldsymbol{\nabla}_p\,\varphi \;+\; \tilde{\gamma}\,[\,\boldsymbol{\nabla}_p(\mathbf{p}\,\varphi) + K_B T\boldsymbol{\Delta}_p\,\varphi\,]\;, \end{aligned}$$

where $\boldsymbol{\Delta}_p$ is the mass-weighted Laplacian in the momenta and φ is a test function [15].

Splitting methods

3.1.2 Some more details about the work of B. Leimkuhler and C. Matthews

The methods that are considered are splitting integrators that decompose the stochastic vector field which drives the Langevin Dynamics into simpler vector fields which can be solved exactly. Nevertheless, the composition method that results cannot be directly related to a stochastic differential equation, but nonetheless the invariant measure associated to the numerical method can be derived, using the Baker-Campbell-Hausdorff (BCH) expansion, as an asymptotic series in two parameters: the stepsize and the coefficient γ^{-1}. The authors stress that their approach focuses on the truncation error of the invariant distribution, thus the greatest benefit would be seen only when the statistical error is well controlled. Furthermore, they observe that the least biased scheme from the point of view of the error introduced in configurational sampling is also as efficient as the alternatives and the most robust with respect the variation of parameters (stepsize and γ^{-1}), thus they emphasise that there is no price for the improvement in accuracy.

Appendix A

Elements of Probability Theory and Stochastic Processes

We introduce some basic definition and results from probability theory and stochastic processes. This material can be found in [14, 16].

A.1 Probability Theory

Definition A.1.1 (**Sample Space**)**.** The set of all events, such as all the possible outcomes of an experiment, is called sample space and is denoted by Ω.

Definition A.1.2 (σ**-algebra**)**.** A collection $\mathscr{F}$ of subset of Ω is called σ-algebra on Ω if[1]

(i) $\Omega \in \mathscr{F}$;
(ii) $F \in \mathscr{F} \implies F^c \coloneqq \Omega \setminus F \in \mathscr{F}$;
(iii) $F_n \in \mathscr{F}$, $n \in \mathbb{N} \implies \bigcup_{n\in\mathbb{N}} F_n \in \mathscr{F}$.

Intuitively, the σ-algebra $\mathscr{F}$ contains all the useful information that is available about the random experiment that we are performing. One of its elements is called a $\mathscr{F}$-measurable subset of Ω or *event*.

Definition A.1.3 (**Measurable Space**)**.** A pair $(\Omega, \mathscr{F})$, where Ω is a set and $\mathscr{F}$ is a σ-algebra on Ω, is called measurable space.

Definition A.1.4 (**Probability Measure**)**.** A probability measure $\mathbb{P}$ on the measurable space $(\Omega, \mathscr{F})$ is a function

$$\mathbb{P}\colon \mathscr{F} \to [0,1];\ F \mapsto \mathbb{P}(F);$$

satisfying

[1] We use the following convention:

$$\mathbb{N} \coloneqq \{1,2,3,\dots\} \subseteq \{0,1,2,3,\dots\} =: \mathbb{Z}^+ \quad \text{and} \quad \mathbb{R}^+ \coloneqq [0,+\infty)\,.$$

(i) $\mathbb{P}(\Omega) = 1$;
(ii) if $F_n \in \mathscr{F}$ with $F_n \bigcap F_m = \emptyset$, $n \neq m$ and $n, m \in \mathbb{N}$, then

$$\mathbb{P}\left(\bigcup_{n\in\mathbb{N}} F_n\right) = \sum_{n\in\mathbb{N}} \mathbb{P}\left(F_n\right) .$$

.

Definition A.1.5 (**Probability Space**)**.** A probability space is a triple $(\Omega, \mathscr{F}, \mathbb{P})$ comprising a set Ω, a σ-algebra $\mathscr{F}$ of subsets of Ω and a probability measure $\mathbb{P}$ on the measurable space $(\Omega, \mathscr{F})$.

Since we are usually interested in the consequence of the outcomes of an experiment, we introduce the notion of random variable that is intuitively a function of the outcomes. We do it following the Williams [16], through the definition of measurable function from $(\Omega, \mathscr{F}, \mathbb{P})$ to $(\mathbb{R}, \mathscr{B}(\mathbb{R}))$[2].

Definition A.1.6 (**$\mathscr{F}$-measurable function**)**.** Let $(\Omega, \mathscr{F}, \mathbb{P})$ and $(\mathbb{R}, \mathscr{B}(\mathbb{R}))$ be two measurable spaces. The function $\psi\colon \Omega \to \mathbb{R}$ is called $\mathscr{F}$-measurable function if

$$\psi^{-1}(B) \coloneqq \{\omega \in \Omega\colon \psi(\omega) \in B\} \in \mathscr{F}, \quad \forall\, B \in \mathscr{B}(\mathbb{R}) .$$

Definition A.1.7 (**Random Variable**)**.** Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space. A random variable is $\mathscr{F}$-measurable function, thus

$$X\colon \Omega \to \mathbb{R}, \quad X^{-1}\colon \mathscr{B}(\mathbb{R}) \to \mathscr{F} .$$

Every random variable, carried by some probability triple $(\Omega, \mathscr{F}, \mathbb{P})$, induces a probability measure $\mathbb{P}_X$ on $(\mathbb{R}, \mathscr{B}(\mathbb{R}))$:

$$\begin{aligned} \mathbb{P}_X\colon \mathscr{B}(\mathbb{R}) &\to [0,1] \\ B &\mapsto \mathbb{P}_X(B) \coloneqq \mathbb{P}(X \in B) = \mathbb{P}(X^{-1}(B)) . \end{aligned}$$

The measure $\mathbb{P}_X$ is called *distribution* (or sometimes the *law*) of X. We can now define the *distribution function* of X:

$$\begin{aligned} F_X\colon \mathbb{R} &\to [0,1] \\ x &\mapsto F_X(x) \coloneqq \mathbb{P}\left(\{\omega \in \Omega\colon X(\omega) \leq x\}\right) = \mathbb{P}\left(X \leq x\right) . \end{aligned}$$

This function F_X indicates the likelihood that the random variable X is in $(-\infty, x]$.

Definition A.1.8 (**Probability density function**)**.** Let X a random variable and $f : \mathbb{R} \to \mathbb{R}$ a function such that

$$f(x) \geq 0 \quad \forall\, x \in \mathbb{R} \qquad \text{and} \qquad \int_{\mathbb{R}} f(x)\,\mathrm{d}x = 1 .$$

We say that X has a *probability density function* (pdf) f_X if

$$\mathbb{P}\left(X \in B\right) = \int_B f_X(x)\,\mathrm{d}x \qquad \forall\, B \in \mathscr{B}(\mathbb{R})$$

[2]We will use the notation $\mathscr{B}(\mathbb{R})$ to denote the Borel σ-algebra of $\mathbb{R}$, the smallest σ-algebra containing all open sets of $\mathbb{R}$.

Definition A.1.9 (**Expected value**). Let $\mathcal{L}^1(\Omega, \mathscr{F}, \mathbb{P})$ be the space of integrable function, see [16, p. 54]. We define the *expectation* $\mathbb{E}[X]$ of a random variable $X \in \mathcal{L}^1(\Omega, \mathscr{F}, \mathbb{P})$ by

$$\mathbb{E}[X] \coloneqq \int_\Omega X \, \mathrm{d}\mathbb{P} \equiv \int_\Omega X(\omega) \, \mathbb{P}(\mathrm{d}\omega) = \int_{\mathbb{R}} x \, \mathbb{P}_X(\mathrm{d}x) \,,$$

In the last equality we changed variable $x = X(\omega)$ and used the definition of distribution of X, that is $\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B))$.

Proposition A.1.10. Let $h : \mathbb{R} \to \mathbb{R}$ a Borel measurable function and $X \in \mathcal{L}^1(\Omega, \mathscr{F}, \mathbb{P})$ a random variable.
Then $h(X) \in \mathcal{L}^1(\Omega, \mathscr{F}, \mathbb{P})$ if and only if $h \in \mathcal{L}^1(\Omega, \mathscr{B}, \mathbb{P}_X)$ and then

$$\mathbb{E}[h(X)] = \int_{\mathbb{R}} h(x) \, \mathbb{P}_X(\mathrm{d}x) \,.$$

Furthermore, if f_X is a pdf of X and

$$\mathbb{E}[|h(X)|] < \infty \iff \int_{\mathbb{R}} |h(x)| \, f_X(x) \, \mathrm{d}x < \infty \,,$$

then

$$\mathbb{E}[h(X)] = \int_{\mathbb{R}} h(x) \, f_X(x) \, \mathrm{d}x \,.$$

Proof. See Williams [16, pp. 68-69]. □

We now give a central definition of modern probability, that is a fundamental theorem (Kolmogorov, 1933).

Theorem A.1.11 (**Conditional expectation**). Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space, $X \in \mathcal{L}^1(\Omega, \mathscr{F}, \mathbb{P})$ a random variable and $\mathscr{G}$ a sub-σ-algebra of $\mathscr{F}$, $\mathscr{G} \subseteq \mathscr{F}$. Then there exists a random variable $Y \in \mathcal{L}^1(\Omega, \mathscr{G}, \mathbb{P})$, that is $\mathscr{G}$-measurable and $\mathbb{E}[|Y|] < \infty$, such that

$$\int_G Y \mathrm{d}\mathbb{P} = \int_G X \mathrm{d}\mathbb{P} \,, \quad \forall \, G \in \mathscr{G} \,.$$

Moreover, if $\tilde{Y}$ is another random variable that has the same properties as Y, then $\tilde{Y} = Y$ *a.s.* (almost surely), i.e. $\mathbb{P}(\tilde{Y} = Y) = 1$.
Therefore, Y with these properties is called **a version of the conditional expectation** $\mathbb{E}[X|\mathscr{G}]$ of X given $\mathscr{G}$, and we write $Y = \mathbb{E}[X|\mathscr{G}]$ *a.s.*

Proof. See Williams [16, pp. 85-86]. □

Generally one identifies several versions and speaks of the conditional expectation $\mathbb{E}[X|\mathscr{G}]$, but we should think about it "*a.s.*".

Intuitively, we have already said that the σ-algebra contains all the useful information that is available on an experiment, while a random variable is a function of its outcomes. In the case the only information we have available regarding which sample point ω has been chosen is the set of values $Z(\omega)$ for every $\mathscr{G}$-measurable random variable Z, i.e. the information available is given by the σ-algebra generated by Z, $\sigma(Z) = \mathscr{G}$. Then the random variable $Y(\omega) = \mathbb{E}[X|\mathscr{G}](\omega)$ is the expected value of $X(\omega)$ given this information. The "*a.s.*" ambiguity in the definition is something with which, in general, one must live.

Definition A.1.12 (Conditional expectation as least-squares-best predictor). Let $X \in \mathcal{L}^1(\Omega, \mathscr{F}, \mathbb{P})$ be a random variable such that $\mathbb{E}\left[X^2\right] < \infty$. Then the conditional expectation $Y = \mathbb{E}[X|\mathscr{G}]$ is a version of the orthogonal projection of X onto $\mathcal{L}^2(\Omega, \mathscr{G}, \mathbb{P})$.
Hence, Y is the least-squares-best $\mathscr{G}$-measurable predictor of X: amongst all $\mathscr{G}$-measurable functions, i.e. amongst all predictors which can be computed from the available information, Y minimize $\mathbb{E}\left[(Y-X)^2\right]$ [16, p. 85].

In other words, this definition says that $Y = \mathbb{E}[X|\mathscr{G}]$ is a version of the orthogonal projection of $X \in \mathcal{L}^2(\Omega, \mathscr{F}, \mathbb{P})$ onto $\mathcal{L}^2(\Omega, \mathscr{G}, \mathbb{P})$ if

$$\mathbb{E}\left[(X-Y)^2\right] \leq \mathbb{E}\left[(X-Z)^2\right],$$

$\forall\, Z$ $\mathscr{G}$-measurable such that $\mathbb{E}\left[Z^2\right] < \infty$.
For a list of properties of the conditional expectation see [16, p. 88].

Proposition A.1.13. Let $X, Z \in \mathcal{L}^1(\Omega, \mathscr{F}, \mathbb{P})$ be two random variables which have a *joint probability density function* $f_{X,Z}(x,z)$. Then

$$f_Z(z) = \int_{\mathbb{R}} f_{X,Z}(x,z)\mathrm{d}x$$

acts as pdf for Z.
Define the *elementary conditional pdf* $f_{X|Z}$ of X given Z via:

$$f_{X|Z}(x|z) \coloneqq \begin{cases} f_{X,Z}(x,z)/f_Z(z) & \text{if } f_Z(z) \neq 0 \\ 0 & \text{otherwise}. \end{cases},$$

Let h be a Borel function on $\mathbb{R}$ such that $\mathbb{E}[|h(X)|] < \infty$. Set

$$g(z) \coloneqq \int_{\mathbb{R}} h(x) f_{X|Z}(x|z)\,\mathrm{d}x\,,$$

then $Y \coloneqq g(Z)$ is a version of the conditional expectation of $h(X)$ given $\sigma(Z)$, $\mathbb{E}[h(X)|\,\sigma(Z)]$.

Proof. We can write $\sigma(Z) = \left\{A = Z^{-1}(B)\,,\ \text{for}\, B \in \mathscr{B}(\mathbb{R})\right\} \subseteq \mathscr{F}$ where

$$Z^{-1}(B) = \{\omega \in \Omega \colon Z(\omega) \in B\} \in \mathscr{F} \quad \forall\, B \in \mathscr{B}(\mathbb{R})\,.$$

Hence, we must show that $\mathbb{E}[h(X)I_B(Z)] = \mathbb{E}[g(Z)I_B(Z)]$ where $I_B(Z)$ is the indicator function of B:

$$I_B(Z) \coloneqq \begin{cases} 1 & \text{if } Z \in B\,, \\ 0 & \text{otherwise}. \end{cases}$$

$$\begin{aligned} \mathbb{E}[h(X)I_B(Z)] &= \iint h(x)I_B(z)f_{X,Z}(x,z)\mathrm{d}x\mathrm{d}z = \\ &= \iint h(x)I_B(z)f_{X|Z}(x|z)\,f_Z(z)\mathrm{d}x\mathrm{d}z = \\ &= \int_{\mathbb{R}} g(z)I_B(z)f_Z(z)\mathrm{d}z = \int_B g(z)f_Z(z)\mathrm{d}z = \mathbb{E}[g(Z)I_B(Z)] \end{aligned}$$

□

A.2 Stochastic Processes

Appendix B

Liouville Space

In classical mechanics, the phase-space[1] $X = \mathbb{R}^{6N}$ is the space of all possible states of a physical system composed by N particles. The state is characterized by the instantaneous positions and momenta of the N particles of the system

$$\boldsymbol{\gamma} = \left(\boldsymbol{\gamma}_{\mathbf{r}}, \boldsymbol{\gamma}_{\mathbf{p}}\right) = (\mathbf{r}_1, \ldots, \mathbf{r}_N, \mathbf{p}_1, \ldots, \mathbf{p}_N) \; . \tag{B.1}$$

Let $\mathcal{H} : X \to \mathbb{R}$ the Hamiltonian of the system and we define a probability measure of finding the system in state $\boldsymbol{\gamma}$ as

$$\mathbb{P}(\mathrm{d}\boldsymbol{\gamma}) = \rho(\boldsymbol{\gamma})\mathrm{d}\boldsymbol{\gamma} \, ,$$

where $\rho(\boldsymbol{\gamma})$ can be any equilibrium ensemble pdf. For example, in the canonical ensemble $\rho(\boldsymbol{\gamma}) = \mathcal{Z}^{-1} e^{-\beta\mathcal{H}(\boldsymbol{\gamma})}$ where $\beta = (K_B T)^{-1} > 0$ such that $e^{-\beta\mathcal{H}(\boldsymbol{\gamma})}$ is integrable with respect to the Lebesgue measure on X and $\mathcal{Z}$ is the canonical partition function, $\mathcal{Z} = \int_X e^{-\beta\mathcal{H}(\boldsymbol{\gamma})}\mathrm{d}\boldsymbol{\gamma}$.

The evolution of a state is obtained in according to the canonical equations of motion as

$$\begin{aligned} \frac{\mathrm{d}\boldsymbol{\gamma}}{\mathrm{d}t} &= \sum_{i=1}^{N} \left[\frac{\partial \boldsymbol{\gamma}}{\partial \mathbf{r}_i} \cdot \dot{\mathbf{r}}_i + \frac{\partial \boldsymbol{\gamma}}{\partial \mathbf{p}_i} \cdot \dot{\mathbf{p}}_i \right] = \\ &= \sum_{i=1}^{N} \left[\frac{\partial \boldsymbol{\gamma}}{\partial \mathbf{r}_i} \cdot \frac{\partial \mathcal{H}}{\partial \mathbf{p}_i} - \frac{\partial \boldsymbol{\gamma}}{\partial \mathbf{p}_i} \cdot \frac{\partial \mathcal{H}}{\partial \mathbf{r}_i} \right] \equiv \{\boldsymbol{\gamma}, \mathcal{H}\} \end{aligned} \tag{B.2}$$

where $\{\boldsymbol{\gamma}, \mathcal{H}\}$ is Poisson bracket of $\boldsymbol{\gamma}$ with the Hamiltonian $\mathcal{H}$. The equation obtained indicates that $\{\boldsymbol{\gamma}, \mathcal{H}\}$ is a generator of the time evolution of $\boldsymbol{\gamma}(t)$.

[1] Mathematically, the space of configuration might be defined by a *manifold* M (either finite or infinite dimensional), and for each position $\mathbf{r} \in M$ in that space, the momentum $\mathbf{p}$ of the system would take values in the cotangent space $T^*_{\mathbf{r}}M$ of that space. This since the velocity $\dot{\mathbf{r}}$ naturally lives in the tangent space $T_{\mathbf{r}}M$, but the momentum, for its definition, lives naturally in the cotangent space $T^*_{\mathbf{r}}M$:

$$\mathbf{p} \coloneqq \frac{\partial L(\mathbf{r}, \dot{\mathbf{r}})}{\partial \dot{\mathbf{r}}},$$

where $L(\mathbf{r}, \dot{\mathbf{r}})$ is the Lagrangian of the system. Therefore the phase-space is naturally represented by the *cotangent bundle*

$$T^*M \coloneqq \{(\mathbf{r}, \mathbf{p}) : \; \mathbf{r} \in M, \, \mathbf{p} \in T^*_{\mathbf{r}}M\} \; ,$$

which comes with a canonical *symplectic form* $\omega \coloneqq \mathrm{d}\mathbf{p} \wedge \mathrm{d}\mathbf{r}$.

The Poisson bracket allows us to introduce an operator L called *Liouvillian* or *Liouville operator*

$$iL \coloneqq \{\ldots, \mathcal{H}\} = \sum_{i=1}^{N} \left[\frac{\partial \mathcal{H}}{\partial \mathbf{p}_i} \cdot \frac{\partial}{\partial \mathbf{r}_i} - \frac{\partial \mathcal{H}}{\partial \mathbf{r}_i} \cdot \frac{\partial}{\partial \mathbf{p}_i} \right] \tag{B.3}$$

where $i = \sqrt{-1}$.

The equation (B.2) can be rewrite as $\mathrm{d}\boldsymbol{\gamma}/\mathrm{d}t = iL\,\boldsymbol{\gamma}$, which can be formally resolved as

$$\boldsymbol{\gamma}(t) = e^{iLt}\,\boldsymbol{\gamma}(0)\,,$$

where the operator $exp(iLt)$ is know as the *classical propagator* and defines a mapping in phase-space.

Let $A : X \to \mathbb{R}$ a *phase functions* satisfying $A \in L^2(X, \mathbb{P})$, where $L^2(X, \mathbb{P})$ is the Lebesgue space of square-summable functions and where functions are identified if and only if they differ only on a set of measure zero. If $L^2(X, \mathbb{P})$ comes with the usual scalar product

$$(A, B) = \int_X A(\boldsymbol{\gamma}) B(\boldsymbol{\gamma})\, \mathbb{P}(\mathrm{d}\boldsymbol{\gamma}) = \int_X A(\boldsymbol{\gamma}) B(\boldsymbol{\gamma})\, \rho(\boldsymbol{\gamma})\, \mathrm{d}\boldsymbol{\gamma}\,, \tag{B.4}$$

then $L^2(X, \mathbb{P})$ is a Hilbert space[2]. It is simple to show that the rate of change of A is given by $\mathrm{d}A/\mathrm{d}t = iL\,A$, which formal solution is given by

$$A(\boldsymbol{\gamma}, t) = e^{iLt}\,A(\boldsymbol{\gamma}, 0)\,. \tag{B.5}$$

The Hilbert space just defined is often called *Liouville space* because the Liouvillian generates the motion in this space.

In a similar way to quantum mechanics, we can now show that the Liouvillian L is a linear Hermitian operator, that is $L^\dagger = L$ where $L^\dagger$ is Hermitian conjugate of L, and thereby the *classical propagator* is a unitary operator in the phase-space.

We can now define the *time correlation function* $C(t)$ as

$$C(t) \coloneqq \langle A(0)\, A(t) \rangle = \int_X A(\boldsymbol{\gamma}, 0) A(\boldsymbol{\gamma}, t)\, \rho(\boldsymbol{\gamma}) \mathrm{d}\boldsymbol{\gamma}\,, \tag{B.6}$$

where $\langle \ldots \rangle$ denotes an ensemble average. This equation can be expressed as

$$C(t) = \int_X A(\boldsymbol{\gamma})\, e^{iLt} A(\boldsymbol{\gamma})\, \rho(\boldsymbol{\gamma}) \mathrm{d}\boldsymbol{\gamma} = \left(A(\boldsymbol{\gamma}), e^{iLt} A(\boldsymbol{\gamma})\right) = (A(0), A(t))\,,$$

thus we obtained

$$\langle A(0)\, A(t) \rangle = (A(0), A(t))\,.$$

[2] In general, let X is a set and μ is a countably additive measure on the σ-algebra of subsets of X. The Hilbert space $L^2(X, \mu)$ is the space of complex-valued measurable functions such that

$$\int_X |f|^2 \mathrm{d}\mu < \infty,$$

where functions are identified if and only if they differ only on a set of measure zero and where the inner product of a pair of functions f, g in $L^2(X, \mu)$ is defined as

$$(f, g) = \int_X f\, g^*\, \mathrm{d}\mu.$$

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